

Performance Testing

Date	30 June 2026
Team ID	
Project Name	Rising Waters – AI-Powered Flood Risk Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Testing
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Flood Risk Prediction Module (/predict) and Risk Probability Analysis
Test Environment	Local Flask Server (Windows 11, Python 3.x)

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid hydrological data for flood risk prediction	10	60	Flood risk level and risk probability should be displayed correctly

2	Multiple users requesting flood risk predictions simultaneously	25	120	System should respond without noticeable delay
3	Submit incomplete or invalid input values	10	60	System should validate inputs and display an error message
4	Continuous prediction requests	50	180	Application should remain stable and responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 sec	1.3 sec	Pass	Fast prediction response
2	Response Time (Max)	< 5 sec	2.9 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	> 10	20 Req/sec	Pass	Stable during testing
4	Error Rate	< 1%	0%	Pass	No errors with valid inputs
5	CPU Utilization	< 80%	45%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	40%	Pass	Efficient memory usage

Step 4: Observations & Analysis

Key Findings

- The application generated flood risk predictions quickly for valid inputs.
- The machine learning model produced predictions consistently during repeated requests.
- The user interface remained responsive throughout testing.
- Resource usage remained within acceptable limits.

Bottlenecks Identified

- Performance may decrease when handling a large number of simultaneous flood prediction requests.
- The application currently depends on historical datasets and does not yet use real-time weather and river monitoring APIs.
- Prediction accuracy may vary if incomplete or outdated hydrological data is provided.

Optimization Steps Taken

- Optimized data preprocessing and feature scaling before prediction.
- Reused the trained machine learning model instead of loading it for every request.
- Simplified Flask request handling to reduce response time.
- Organized the code into reusable modules for easier maintenance and future enhancements.
- Reduced prediction latency by caching frequently used resources and model artifacts.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing results below (e.g., browser testing results, API response screenshots, prediction dashboards, system monitoring outputs).

Screenshot 1

Enter Weather Readings

Provide current meteorological data for a flood-prone district.

Annual Rainfall (mm)

Monsoon Rainfall (mm)

Pre-Monsoon Rainfall (mm)

Post-Monsoon Rainfall (mm)

Cloud Visibility (km)

Predict Flood Risk

Prediction Result

Submit weather readings to receive an instant flood risk classification.

Use Cases

- **Early warning:** Issue evacuation advisories hours in advance.
- **Resource allocation:** Prioritise relief deployment across regions.
- **Model validation:** Compare predictions against historical events.

Screenshot 2

Enter Weather Readings

Provide current meteorological data for a flood-prone district.

Annual Rainfall (mm)

Monsoon Rainfall (mm)

Pre-Monsoon Rainfall (mm)

Post-Monsoon Rainfall (mm)

Cloud Visibility (km)

Predict Flood Risk

Prediction Result

Low Risk

No flood predicted

Flood probability: 0.02%

Low flood risk based on current readings.
Continue routine monitoring during the monsoon season.

Use Cases

- **Early warning:** Issue evacuation advisories hours in advance.
- **Resource allocation:** Prioritise relief deployment across regions.
- **Model validation:** Compare predictions against historical events.